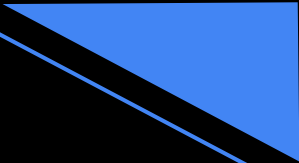


Winning Space Race with Data Science

CHLOE TENG
28 August 2026



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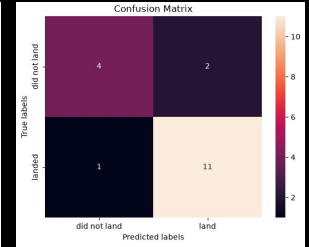
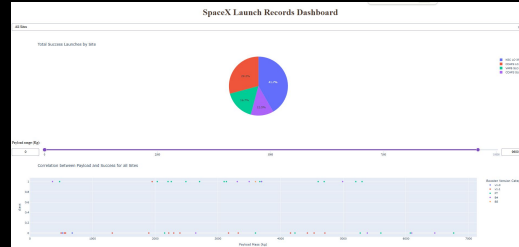
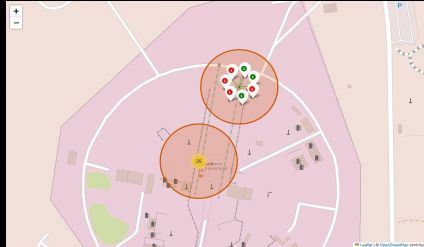
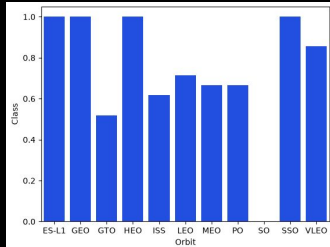


Executive Summary

Summary of Methodologies

This project follows these steps:

1. Data Collection
2. Data Wrangling
3. Exploratory Data Analysis (EDA)
4. Interactive Visual Analytics
5. Predictive Analysis (Classification)



Summary of Results

This project produced the following outputs and visualisations:

1. Exploratory Data Analysis (EDA) Results
2. Geospatial Analytics (Folium Interactive Map)
3. Interactive Dashboard (Plotly Dash)
4. Predictive Analysis of Classification Models

Introduction

SpaceX has significantly reduced the cost of rocket launches by developing reusable Falcon 9 rockets. The company advertises Falcon 9 launches at approximately **\$62 million**, while traditional launch providers may charge **over \$165 million** per launch.

A major factor behind this cost advantage is **SpaceX's** ability to **recover and reuse the Falcon 9 first stage**. However, the success of first-stage landing depends on various mission factors, such as launch site, payload, orbit type, and booster characteristics.

In this project, we analyse historical Falcon 9 launch data and apply machine learning techniques to **predict whether the first stage will successfully land**. The results can help estimate launch costs and provide insights for companies seeking to compete in the commercial space launch market.

1. Can we predict whether the Falcon 9 first stage will successfully land based on historical launch data?
2. Which factors have the greatest impact on first-stage landing success?
3. What launch conditions or configurations provide the highest probability of a successful landing?
4. Can the prediction model help estimate launch costs and support decision-making for companies competing with **SpaceX**?



Methodology Summary

1. Data Collection

- a. Collected Falcon 9 launch data using the **SpaceX REST API**
- b. Performed **web scraping from Wikipedia Falcon 9 launch records** to obtain historical launch information

2. Data Wrangling and Data Processing

- a. Cleaned and transformed the collected data using Pandas
- b. Handled missing values and formatted variables for analysis
- c. **Created the landing outcome class label:** 1 = Successful first-stage landing, 0 = Unsuccessful first-stage landing
- d. Prepared features including launch site, payload mass, orbit type, and booster information

3. Exploratory Data Analysis (EDA)

- a. Used **SQL queries** to filter, aggregate, and analyze the **SpaceX** launch dataset
 - b. Applied **Pandas, Matplotlib, and Seaborn** visualization techniques to explore relationships between launch factors and landing success
- 



Methodology Summary

4. Interactive Visual Analytics

- a. Used **Folium** for geospatial analysis of launch sites and success patterns
- b. Developed an interactive **Plotly Dash dashboard** to analyze launch performance based on launch site and payload characteristics

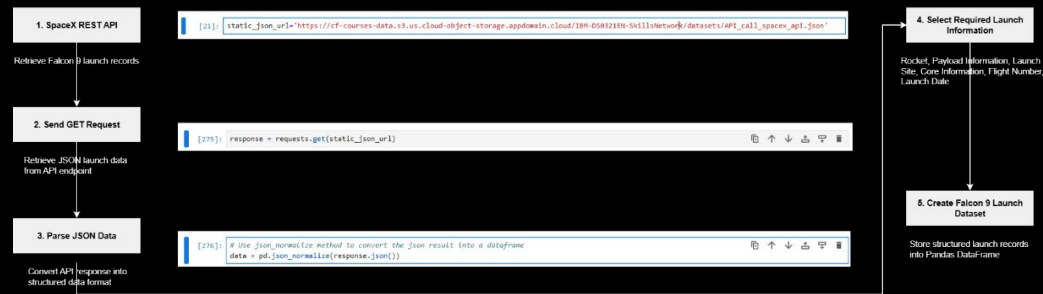
5. Predictive Analysis and Model Evaluation

- c. Applied classification algorithms using **Scikit-Learn**
 - d. Standardized features and split data into training and testing datasets
 - e. Trained multiple classification models: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbor (KNN)
 - f. Used **GridSearchCV** for hyperparameter tuning
 - g. Evaluated models using: Accuracy score and Confusion matrix
- 

Data Collection - SpaceX REST API

Objective: Retrieve structured Falcon 9 launch data from the **SpaceX** REST API, including rocket, payload information, launch site, core information, flight number, and launch date.

Flowchart (See Slide 54 Appendix for full view):



```
[59]: # Lets take a subset of our dataframe keeping only the features we want and the flight number, and date_utc.
data = data[['rocket', 'payloads', 'launchpad', 'cores', 'flight_number', 'date_utc']]

"""We will remove rows with multiple cores because those are falcon rockets with 2 extra rocket boosters and rows that have multiple
payloads in a single rocket."""
data = data[data['cores'].map(len)==1]
data = data[data['payloads'].map(len)==1]

# Since payloads and cores are lists of size 1 we will also extract the single value in the list and replace the feature.
data['cores'] = data['cores'].map(lambda x : x[0])
data['payloads'] = data['payloads'].map(lambda x : x[0])

# We also want to convert the date_utc to a datetime datatype and then extracting the date leaving the time
data['date'] = pd.to_datetime(data['date_utc']).dt.date

# Using the date we will restrict the dates of the launches
data = data[data['date'] <= datetime.date(2020, 11, 11)]
```

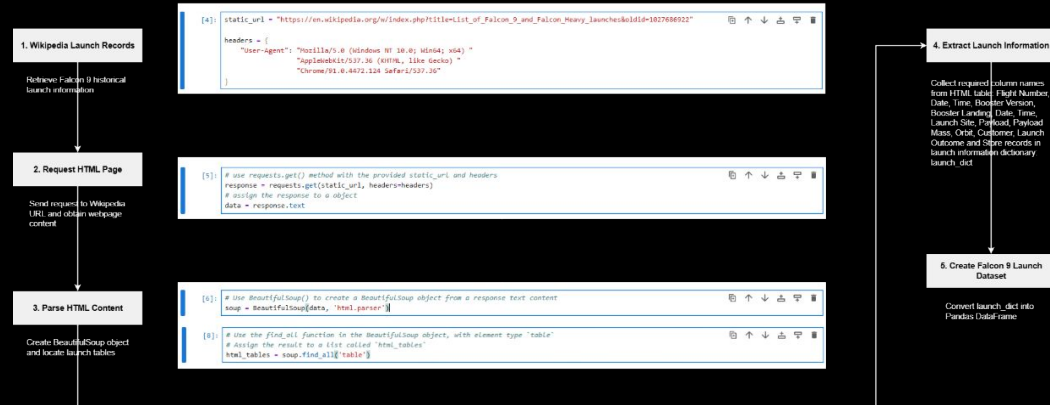
```
[70]: df = pd.read_csv(spacex_url)
```

[GitHub URL of My SpaceX REST API Notebook](#)

Data Collection - Web Scraping

Objective: Collect historical Falcon 9 launch records by scraping launch tables from a Wikipedia page titled "List of Falcon 9 and Falcon Heavy launches".

Flowchart (See Slide 54 Appendix for full view):



```
[10]: column_names = []

# Apply find_all() function with "th" element on first_launch_table and iterate each th element
for row in first_launch_table.find_all("th"):
    name = extract_column_from_header(row)
    # Append the non-empty column name ("if name is not None and len(name) > 0") into a list called column_names
    if (name != None and len(name) > 0):
        column_names.append(name)

[12]: # Let's initiate the launch_dict with each value to be an empty list
launch_dict["Flight No."] = []
launch_dict["Launch site"] = []
launch_dict["Payload"] = []
launch_dict["Payload mass"] = []
launch_dict["Orbit"] = []
launch_dict["Customer"] = []
launch_dict["Launch outcome"] = []
# Added some new columns
launch_dict["Version Booster"] = []
launch_dict["Booster landing"] = []
launch_dict["Date"] = []
launch_dict["Time"] = []

[14]: df = pd.DataFrame([keypd.Series(value) for key, value in launch_dict.items()])
```

[GitHub URL of My Web Scraping Notebook](#)

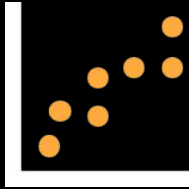

```
[1]: df_gd.read_csv("https://cf-cddcra-data.s3.us-east-1.amazonaws.com/appdevr/cloud/D0M-CDDCRA-1681167497/datasets/dataset.parquet")
```



Exploratory Data Analysis (EDA) - Visualization

01

Scatter Plots

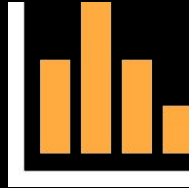


Used to visualize relationships between launch characteristics and landing success (Class = 0 for unsuccessful landing; Class = 1 for successful landing).

- Flight Number vs Launch Site
- Payload Mass vs Launch Site
- Flight Number vs Orbit Type
- Payload Mass vs Orbit Type

02

Bar Charts



Used to compare landing success rates across categorical variables,

- Success Rate vs Orbit Type

03

Line Charts



Used to analyse changes and trends in landing success over time.

- Year vs Landing Success Rate

[GitHub URL of My EDA Visualization Notebook](#)

Exploratory Data Analysis (EDA) - SQL

To extract key insights about the **SpaceX** launch dataset, multiple SQL queries were performed:

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version Fg v1.1
- List the date when the first successful landing outcome in ground pad was achieved
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the boosters which have carried the maximum payload mass
- List the failed landing outcomes in drone ship, their booster versions, and launch site names in 2015
- Rank the count of landing outcomes such as Failure (drone ship) or Success (ground pad) between the date 2010-06-04 and 2017-03-20, in descending order

Geospatial Analysis - Folium Interactive Map

The following steps were taken to visualize the launch data on the map:

1. Mark all launch sites

[GitHub URL of My Interactive Map \(Folium\) Notebook](#)

- a. Initialize the interactive map using **folium.Map** as the base map for displaying **SpaceX** launch locations
- b. Add **folium.Circle** objects to highlight the geographic positions of launch sites
- c. Add **folium.map.Marker** objects with **DivIcon** labels to display launch site names
- d. Add **folium.Popup** objects to display launch site names when users click on the locations

2. Mark successful and failed launches for each site

- a. Add **MarkerCluster** objects to group multiple launches at the same location and improve map readability
- b. Add **folium.map.Marker** objects with **folium.Icon** colours: green for successful landings (Class = 1) and red for unsuccessful landings (Class = 0) to visually distinguish landing outcomes

3. Visualize distances between launch sites and nearby locations

- a. Add **folium.Marker** objects with **DivIcon** labels to display calculated distances between launch sites and nearby locations
- b. Add **folium.PolyLine** objects to draw connections between launch sites and nearby locations

Interactive Dashboard - Plotly Dash

The following plots and interactive components were added to the dashboard to analyze **SpaceX** launch records:

[GitHub URL of My Interactive Dashboard \(Plotly Dash\) Notebook](#)

1. Launch Site Selection

Added a **dcc.Dropdown** component to allow users to select a specific launch site or view all sites. This enables users to dynamically filter dashboard results based on launch site selection.

2. Success Launch Distribution

Added a **plotly.express.pie** pie chart to display launch success distribution across different launch sites. This provides a visual comparison of launch success performance between sites.

3. Payload Range Filtering

Added a **dcc.RangeSlider** component to filter launches based on payload mass range. This allows users to explore launch outcomes for different payload sizes.

Interactive Dashboard - Plotly Dash

4. Payload Mass vs Launch Success

[GitHub URL of My Interactive Dashboard \(Plotly Dash\) Notebook](#)

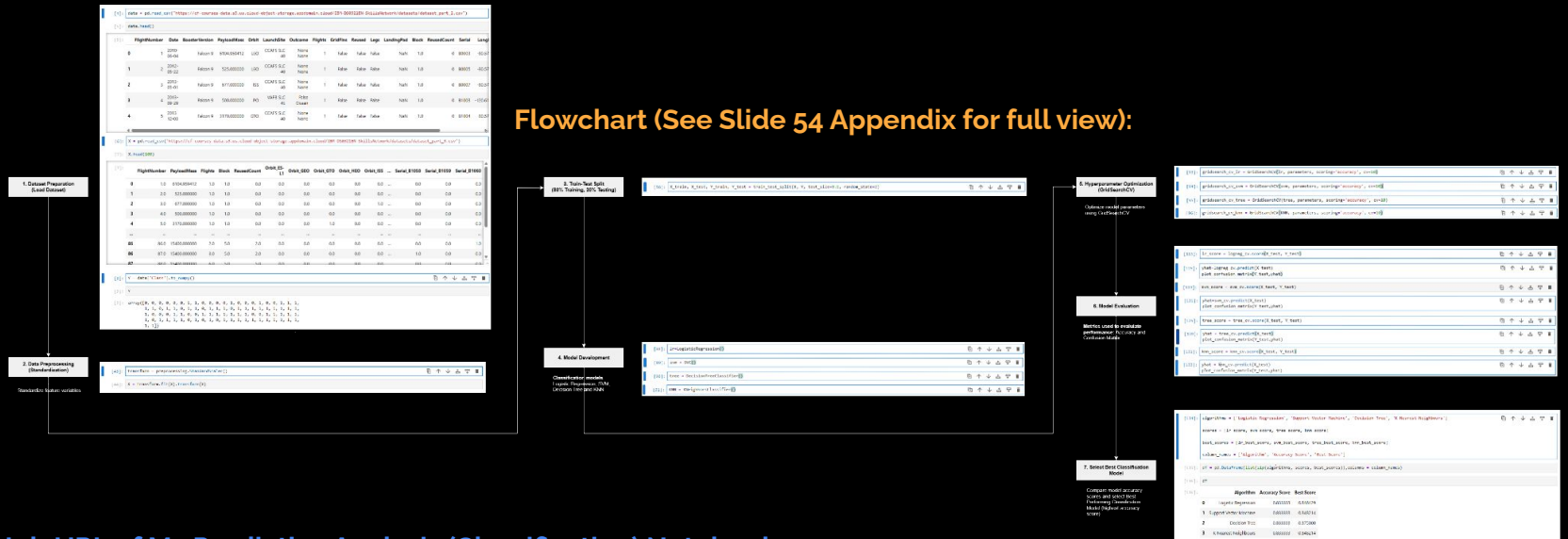
Added a **plotly.express.scatter** plot to visualize the relationship between payload mass and landing success (Class). This helps to identify patterns between payload mass, launch success, and booster version categories.

5. Interactive Dashboard Updates

Added **Dash callback functions** to update charts based on dropdown and slider inputs. This allows users to dynamically explore different launch sites and payload ranges without generating separate charts.

The following steps were performed to develop, optimize, evaluate, and select the best performing classification model:

Flowchart (See Slide 54 Appendix for full view):



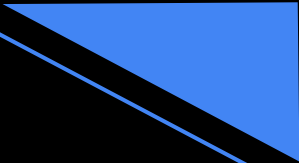
GitHub URL of My Predictive Analysis (Classification) Notebook



Results

- 01** Exploratory Data Analysis (EDA)
- 02** Interactive Analytics
- 03** Predictive Analysis - Classification





Results

01 EDA with Visualization



Flight Number vs Launch Site

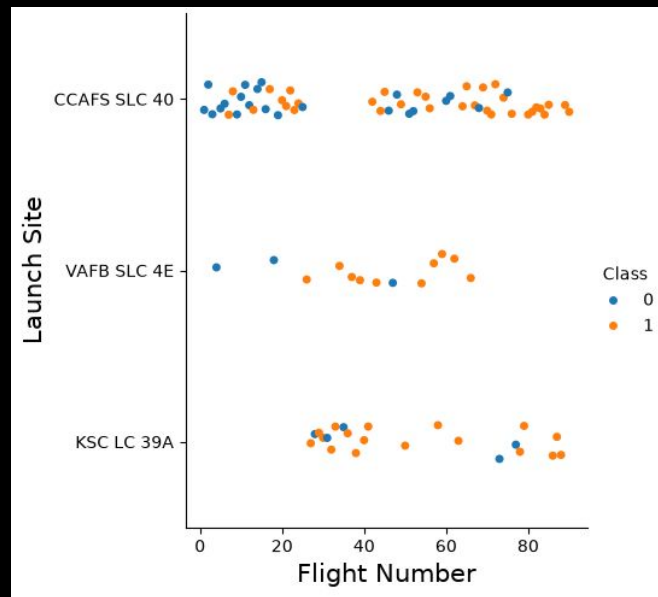
The scatter plot of Flight Number vs Launch Site shows that:

1. **Higher flight numbers have more successful landing outcomes across different launch sites**

Later Falcon 9 flights contain more successful landings (Class = 1), suggesting improved landing performance over time.

2. **CCAFS SLC 40 has the largest number of launches across the dataset**

This site has both early and later missions, with successful and unsuccessful landing outcomes observed across different flight numbers.



Flight Number vs Launch Site

3. VAFB SLC 4E has successful and unsuccessful landing outcomes across different flight numbers

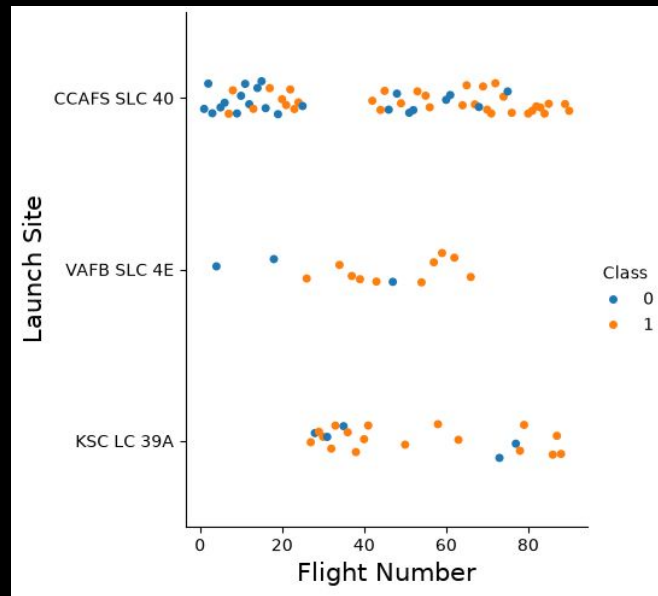
Early flights have more unsuccessful landing outcomes, while later missions mostly have successful landing outcomes.

4. KSC LC 39A mainly has later flight numbers with mostly successful landing outcomes

Most launches from this site occurred at higher flight numbers and have a higher proportion of successful landings.

5. Launch site alone does not show a clear relationship with landing outcomes

Within each launch site, higher flight numbers generally have more successful landing outcomes, suggesting improved landing performance over time.



Payload Mass vs Launch Site

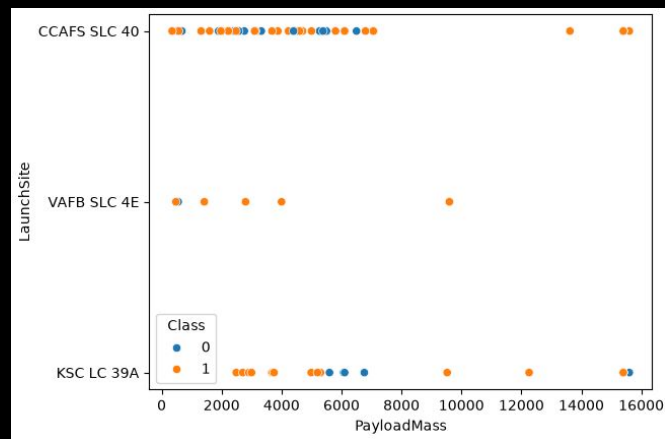
The scatter plot of Payload Mass vs Launch Site shows that:

1. **CCAFS SLC 40 has the largest number of launches across a wide range of payload masses**

Most launches from this site have payload masses below approximately 7,000 kg, with both successful and unsuccessful landing outcomes observed.

2. **Higher payload masses do not show a clear relationship with landing success**

Successful and unsuccessful landing outcomes are observed across different payload ranges, suggesting payload mass alone does not show a clear relationship with landing outcomes.



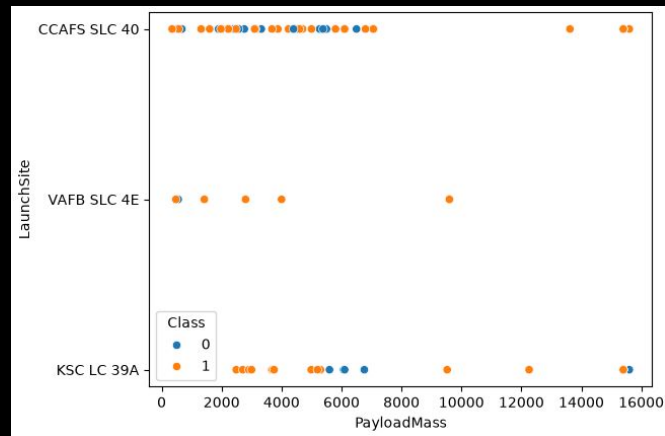
Payload Mass vs Launch Site

3. KSC LC 39A has mainly higher payload mass missions

Most launches from this site have payload masses above approximately 2,500 kg, with the majority having successful landing outcomes.

4. VAFB SLC 4E has fewer launches, with successful and unsuccessful landing outcomes observed across different payload masses

The limited number of observations makes it difficult to identify a strong payload-related trend.

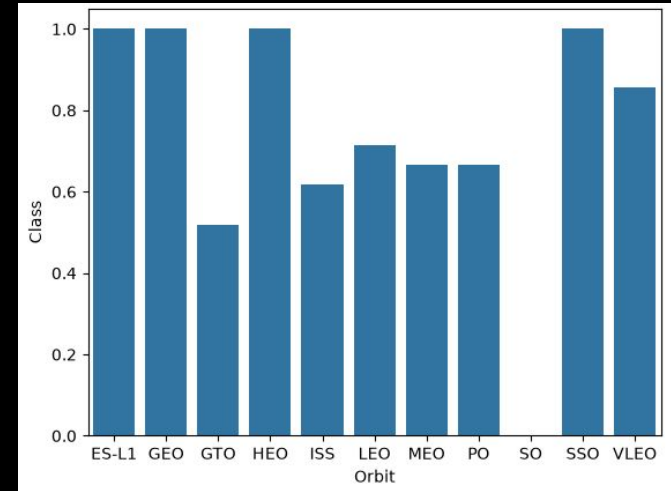


Overall, payload mass alone does not show a clear relationship with landing outcomes.

Success Rate vs Orbit Type

The bar chart of Success Rate vs Orbit Type shows that:

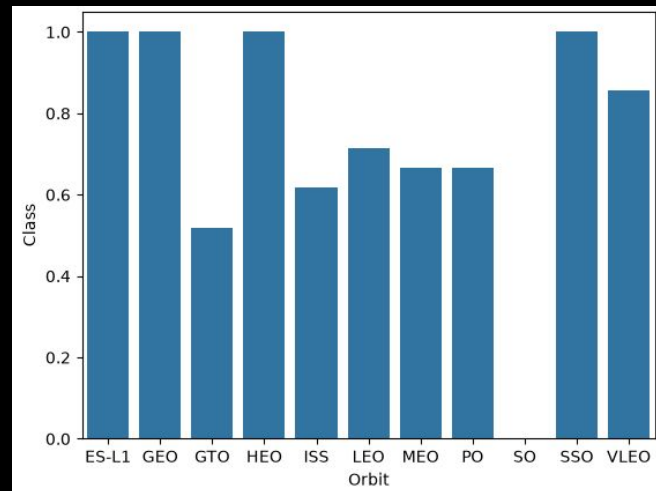
1. **ES-L1, GEO, HEO, and SSO have the highest success rates at 100%**
2. **SO has the lowest success rate at 0%**
3. **VLEO has a high success rate of approximately 85%, while LEO has around 71%**
4. **MEO and PO have similar success rates of approximately 67%, indicating moderate landing performance**



Success Rate vs Orbit Type

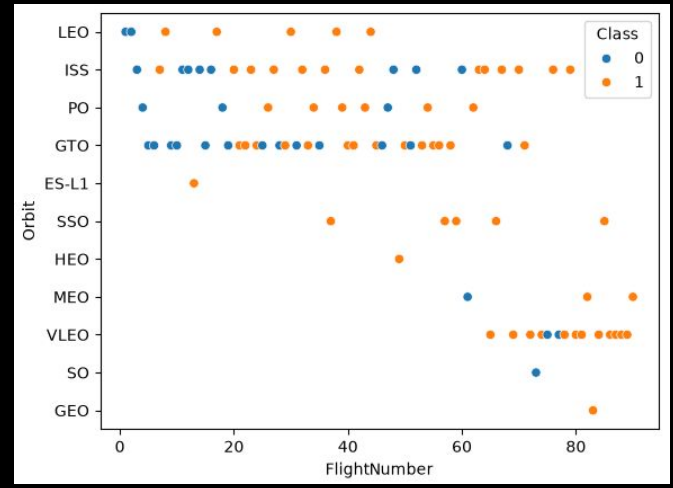
5. GTO and ISS have lower success rates compared with other orbit types, with GTO having approximately 50% success

Overall, landing success varies across orbit types, suggesting that orbit type may be an important factor in predicting Falcon 9 landing outcomes.





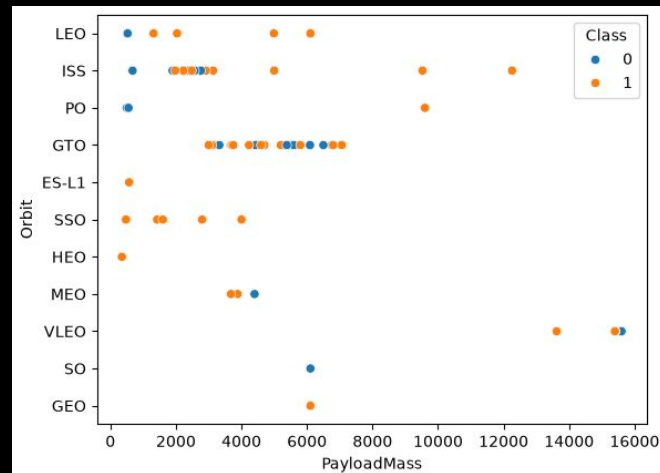
- 



Payload Mass vs Orbit Type

The scatter plot of Payload Mass vs Orbit Type shows that:

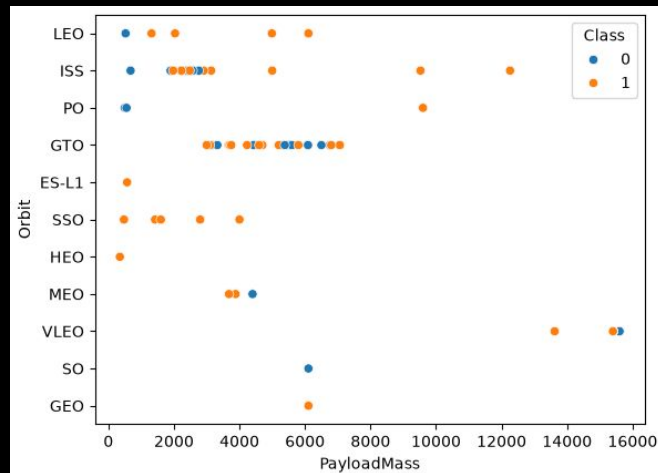
1. **VLEO launches mainly occur at higher payload masses and have mostly successful landing outcomes**
2. **GTO has the largest number of launches across a wide range of payload masses**
3. **LEO and ISS have successful and unsuccessful landing outcomes across different payload ranges**



Payload Mass vs Orbit Type

4. Some orbit types, such as GEO and ES-L1, have limited payload observations, making it difficult to identify a clear trend

Overall, payload mass distribution varies across orbit types, but payload mass alone does not show a clear relationship with landing outcomes.



Launch Success Yearly Trend

The line chart of average Launch Success Yearly Trend shows that:

1. Landing success rate increases after 2013

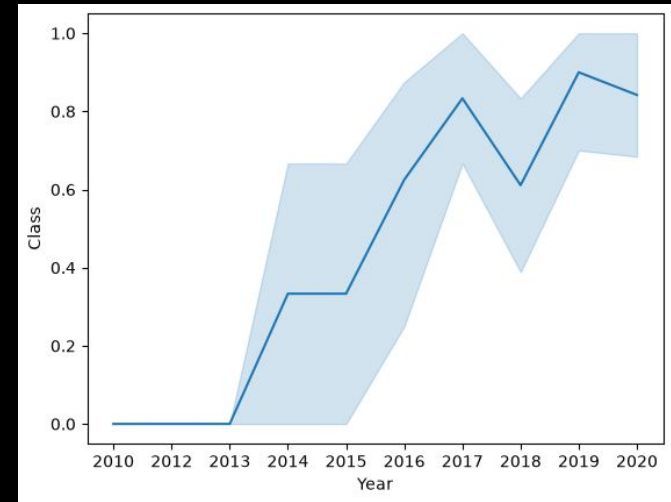
The success rate remains at 0 before 2013, then rises in later years, indicating improved landing performance over time.

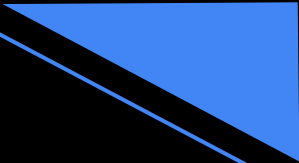
2. Later years achieve higher success rates

Most years after 2016 show success rates above 50%, with the highest values occurring around 2017 and 2019.

3. Yearly success rates fluctuate across different years

The success rate decreases in 2018 before increasing again in 2019, indicating variation in yearly landing outcomes.





Results

01 EDA with SQL

All Launch Site Names

Find the names of the launch sites.

SQL Query:

```
[48]: %sql SELECT DISTINCT LAUNCH_SITE FROM SPACEXTABLE;
```

SQL Query Result:

[48]:	Launch_Site
	CCAFS LC-40
	VAFB SLC-4E
	KSC LC-39A
	CCAFS SLC-40

SQL Explanation:

Using **DISTINCT** returns only unique values from the **LAUNCH_SITE** column of the **SPACEXTABLE** table.

The query returns the different launch locations used in the **SpaceX** launch dataset.

Launch Site Names Begin with 'CCA'

Find 5 records where launch sites begin with 'CCA'.

SQL Query:

```
[49]: %sql SELECT LAUNCH_SITE FROM SPACEXTABLE WHERE LAUNCH_SITE LIKE 'CCA%' LIMIT 5;
```

SQL Query Result:

[49]:

Launch_Site
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

SQL Explanation:

Using **LIKE 'CCA%'** filters the **LAUNCH_SITE** column of the **SPACEXTABLE** table to identify launch sites beginning with 'CCA.', and **LIMIT 5** displays the first five matching records from the **SpaceX** launch dataset.

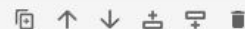
The query returns the first five launch site records matching the specified pattern in the dataset.

Total Payload Mass

Calculate the total payload carried by boosters from NASA.

SQL Query:

```
[50]: %sql SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD_MASS FROM SPACEXTABLE \
      WHERE CUSTOMER = 'NASA (CRS)';
```



SQL Query Result:

[50]: TOTAL_PAYLOAD_MASS
45596

SQL Explanation:

Using **SUM(PAYLOAD_MASS__KG_)** calculates the total payload mass from the **PAYLOAD_MASS__KG_** column of the **SPACEXTABLE** table for records **WHERE** the **CUSTOMER** is 'NASA (CRS)'.

The query returns the total payload mass carried by boosters launched for NASA (CRS) missions in the **SpaceX** launch dataset.

Average Payload Mass by F9 v1.1

Calculate the average payload mass carried by booster version F9 v1.1.

SQL Query:

```
[51]: %sql SELECT AVG(PAYLOAD_MASS_KG_) AS AVERAGE_PAYLOAD_MASS FROM SPACEXTABLE \
      WHERE BOOSTER_VERSION = 'F9 v1.1';
```



SQL Query Result:

[51]: AVERAGE_PAYLOAD_MASS
2928.4

SQL Explanation:

Using **AVG(PAYLOAD_MASS_KG_)** calculates the average payload mass from the **PAYLOAD_MASS_KG_** column of the **SPACEXTABLE** table for records **WHERE** the **BOOSTER_VERSION** is 'F9 v1.1'.

The query returns the average payload mass carried by the F9 v1.1 booster version in the **SpaceX** launch dataset.

First Successful Ground Landing Date

Find the dates of the first successful landing outcome on ground pad.

SQL Query:

```
[52]: %sql SELECT MIN(DATE) AS FIRST_SUCCESSFUL_GROUND_LANDING FROM SPACEXTABLE \
      WHERE Landing_Outcome = 'Success (ground pad)';
```



SQL Query Result:

```
[52]: FIRST_SUCCESSFUL_GROUND_LANDING
      2015-12-22
```

SQL Explanation:

Using **MIN(****DATE)** returns the earliest date from the **DATE** column of the **SPACEXTABLE** table for records **WHERE** the **Landing_Outcome** is 'Success (ground pad)'.

The query returns the first successful ground pad landing date in the **SpaceX** launch dataset.



Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000.

SQL Query:

```
[53]: %sql SELECT BOOSTER_VERSION FROM SPACEXTABLE \
      WHERE (Landing_Outcome = 'Success (drone ship)') \
      AND (PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000);
```



SQL Query Result:

[53]: **Booster_Version**

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

SQL Explanation:

Using the **Landing_Outcome** and **PAYLOAD_MASS__KG_** columns from the **SPACEXTABLE** table identifies records **WHERE** the **Landing_Outcome** is 'Success (drone ship)' and the **PAYLOAD_MASS__KG_** is greater than 4000 and less than 6000.

The query returns the booster versions that successfully landed on a drone ship with payload masses within the specified range in the **SpaceX** launch dataset.



Total Number of Successful and Failure Mission Outcomes

Calculate the total number of successful and failure mission outcomes.

SQL Query:

```
[54]: %sql SELECT MISSION_OUTCOME, COUNT(MISSION_OUTCOME) AS TOTAL_NUMBER FROM SPACEXTABLE GROUP BY MISSION_OUTCOME;
```

SQL Query Result:

[54]:	Mission_Outcome	TOTAL_NUMBER
	Failure (in flight)	1
	Success	98
	Success	1
	Success (payload status unclear)	1

SQL Explanation:

Using **COUNT(MISSION_OUTCOME)** calculates the total number of records for each **MISSION_OUTCOME** category from the **SPACEXTABLE** table.

The query returns the total number of successful and failure mission outcomes in the **SpaceX** launch dataset.

Boosters Carried Maximum Payload

List the names of the booster which have carried the maximum payload mass.

SQL Query:

```
[17]: %sql SELECT DISTINCT(BOOSTER_VERSION) FROM SPACEXTABLE \
      WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

SQL Query Result:

```
[17]: Booster_Version
      F9 B5 B1048.4
      F9 B5 B1049.4
      F9 B5 B1051.3
      F9 B5 B1056.4
      F9 B5 B1048.5
      F9 B5 B1051.4
      F9 B5 B1049.5
      F9 B5 B1060.2
      F9 B5 B1058.3
      F9 B5 B1051.6
      F9 B5 B1060.3
      F9 B5 B1049.7
```

SQL Explanation:

Using **DISTINCT(BOOSTER_VERSION)** returns only unique booster version values from the **BOOSTER_VERSION** column of the **SPACEXTABLE** table **WHERE PAYLOAD_MASS__KG_** equals the maximum payload mass calculated by **MAX(PAYLOAD_MASS__KG_)**.

The query returns the booster versions that carried the maximum payload mass in the **SpaceX** launch dataset.

2015 Launch Records

List the failed landing_outcomes in drone ship, their booster versions, and launch site names in 2015.

SQL Query:

```
[18]: %sql SELECT substr(Date, 6, 2) AS MONTH, Landing_Outcome, BOOSTER_VERSION, LAUNCH_SITE FROM SPACEXTABLE \
WHERE substr(Date, 0, 5) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

SQL Query Result:

[18]:	MONTH	Landing_Outcome	Booster_Version	Launch_Site
	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

SQL Explanation:

Using **substr(Date, 6, 2)** returns the month from the **Date** column of the SPACEXTABLE table for records **WHERE substr(Date, 0, 5)** is '2015' and **Landing_Outcome** is 'Failure (drone ship)'.

The query returns the month, landing outcome, booster version, and launch site for failed drone ship landing records in 2015 from the **SpaceX** launch dataset.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

SQL Query:

```
[19]: %sql SELECT Landing_Outcome, COUNT(Landing_Outcome) AS TOTAL_NUMBER FROM SPACEXTABLE \
      WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' \
      GROUP BY Landing_Outcome \
      ORDER BY TOTAL_NUMBER DESC;
```

SQL Query Result:

[19]:	Landing_Outcome	TOTAL_NUMBER
	No attempt	10
	Success (drone ship)	5
	Failure (drone ship)	5
	Success (ground pad)	3
	Controlled (ocean)	3
	Uncontrolled (ocean)	2
	Failure (parachute)	2
	Precluded (drone ship)	1

SQL Explanation:

Using **COUNT(Landing_Outcome)** calculates the total number of records for each **Landing_Outcome** category from the **SPACEXTABLE** table between '2010-06-04' and '2017-03-20', and **ORDER BY TOTAL_NUMBER DESC** ranks the counts in descending order.

The query returns the landing outcome counts ranked from highest to lowest for the specified date range in the **SpaceX** launch dataset.



Results

02 Launch Sites Proximity Analysis - Folium Interactive Map

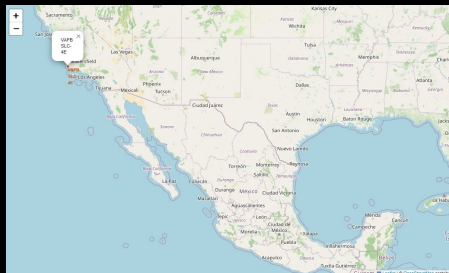


All Launch Sites On The Map

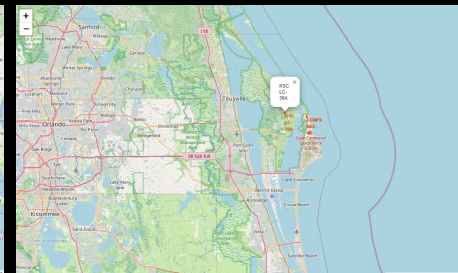
The map screenshots shows the locations of four **SpaceX** launch sites across the United States:

1. **CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A are located close to each other on Florida's east coast, near Cape Canaveral and Kennedy Space Center. While VAFB SLC-4E is located separately on the west coast in California**
2. **All launch sites shown are near coastal areas, which is important for rocket launch safety and access to open ocean flight paths**

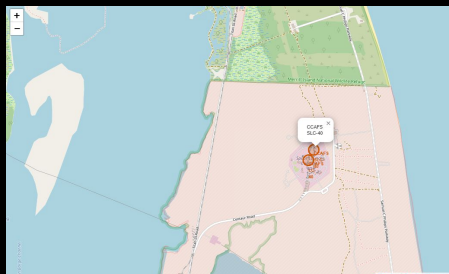
VAFB SLC-4E
(34.632834, -120.610745)



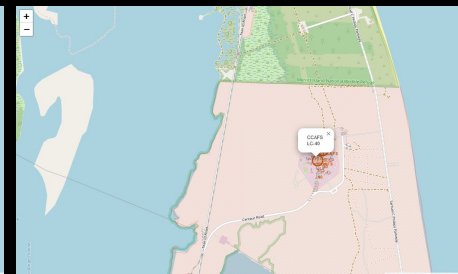
KSC LC-39A
(28.573255, -80.646895)



CCAFS SLC-40
(28.563197, -80.576820)



CCAFS LC-40
(28.562302, -80.577356)

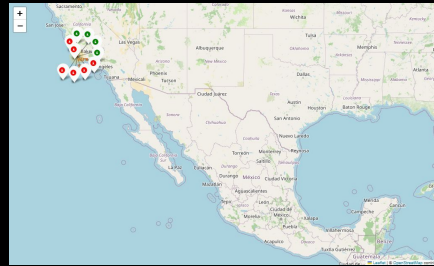


All Success / Failed Launches For Each Site On The Map

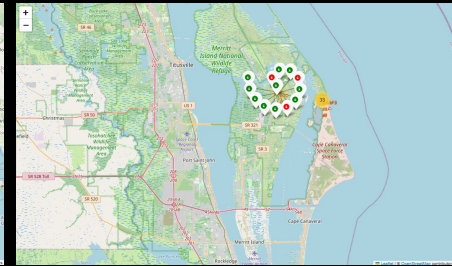
The map screenshots shows successful and failed launches at four **SpaceX** launch sites using color-coded markers, green and red:

1. **KSC LC-39A has more green markers, suggesting a relatively higher launch success rate**
2. **CCAFS LC-40 and CCAFS SLC-40 have more red markers, suggesting lower launch success rates**
3. **VAFB SLC-4E has both green and red markers, showing mixed launch outcomes**

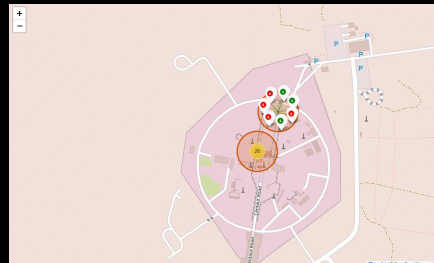
VAFB SLC-4E
(34.632834, -120.610745)



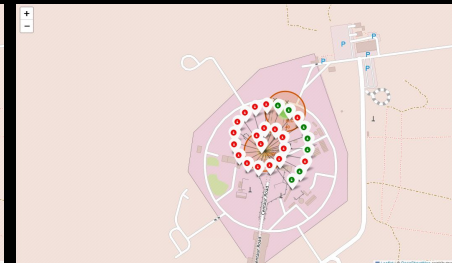
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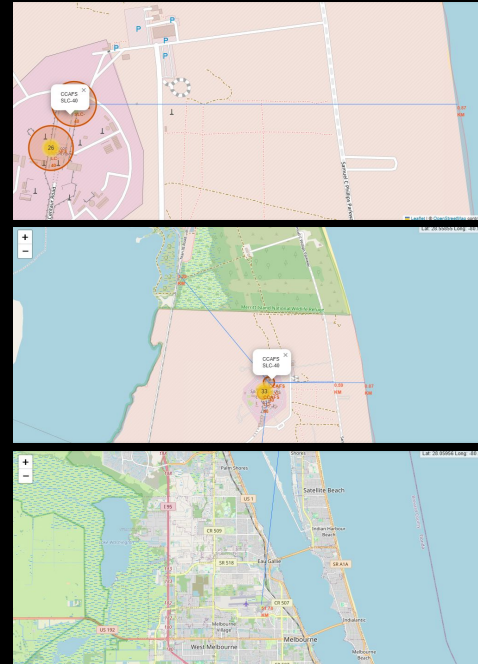
CCAFS LC-40
(28.562302, -80.577356)

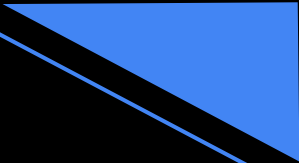


Proximity Of A Selected Launch Site To Other Points of Interest

The map screenshots show the approximate distance from the **CCAFS SLC-40** launch site to nearby points of interest using blue distance lines, including the coastline, highway, railway, and city locations:

1. The coastline is 0.87 km due east of the launch site
2. The nearest highway is 0.59 km away
3. The nearest railway is 1.29 km away
4. The nearest city is 51.74 km away





Results

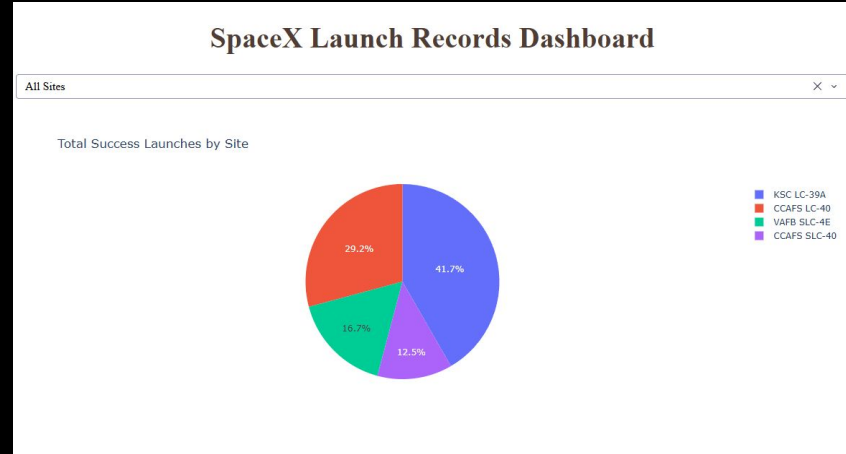
02 interactive Dashboard - Plotly Dash



Total Success Launches For All Sites

The pie chart of Total Success Launches For All Sites shows that:

1. **KSC LC-39A had the most successful launches, accounting for 41.7% of the total**
2. **CCAFS LC-40 had the second most successful launches, accounting for 29.2% of the total**
3. **VAFB SLC-4E accounted for 16.7% of the total successful launches**
4. **CCAFS SLC-40 had the fewest successful launches, accounting for 12.5% of the total**

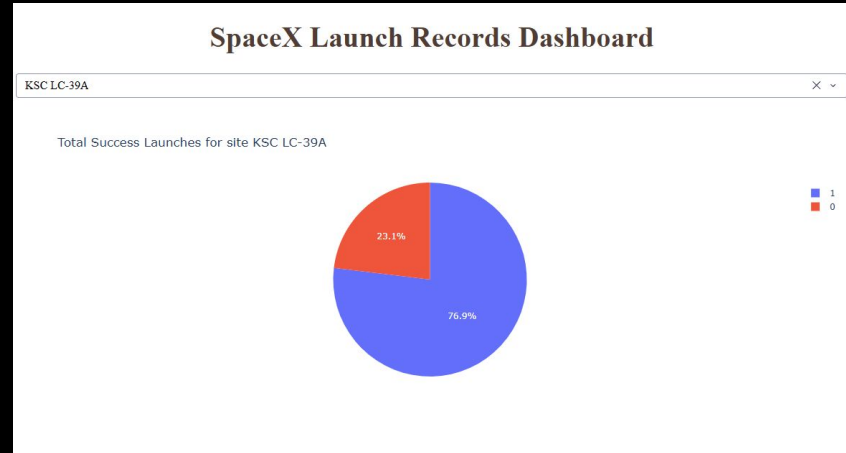


Overall, KSC LC-39A was the most successful launch site, accounting for 41.7% of total successful launches, while CCAFS SLC-40 had the lowest share among the four sites shown.

Launch Site With Highest Launch Success Ratio

The pie chart of Launch Site With Highest Launch Success Ratio shows that:

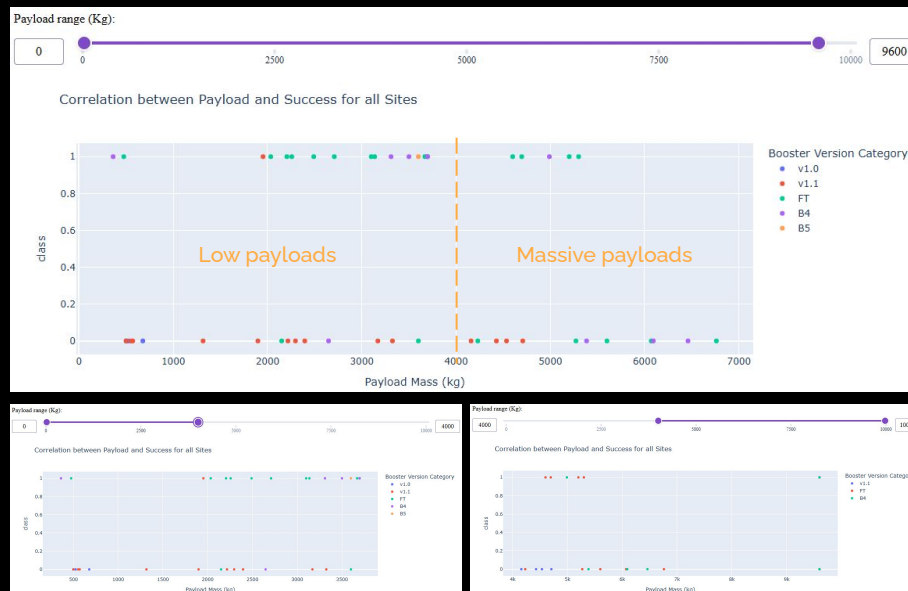
KSC LC-39A had the highest launch success ratio among all launch sites, with 76.9% successful launches and 23.1% unsuccessful launches.



Payload Mass vs Launch Outcome For All Sites

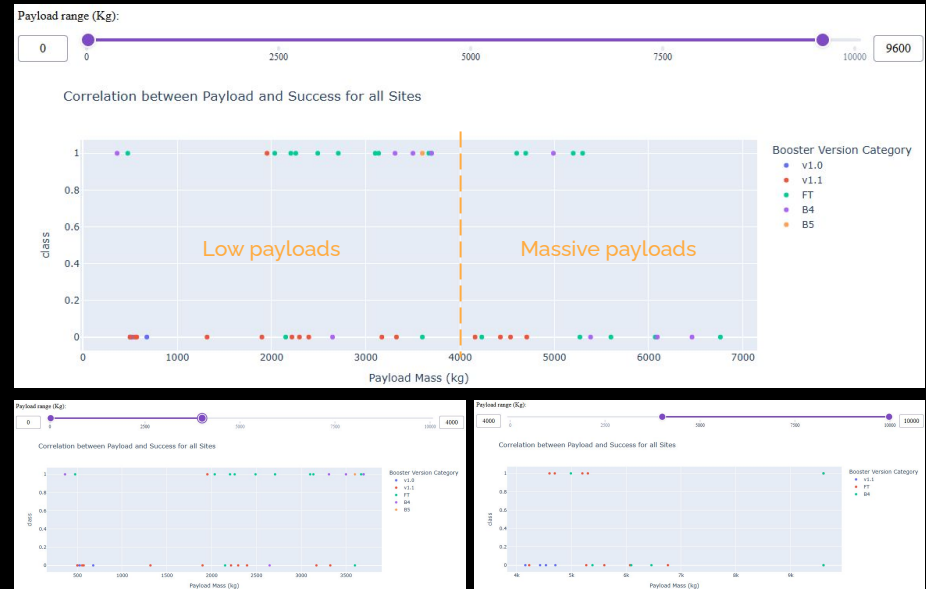
The scatter plots of Payload Mass vs Launch Outcome For All Sites, with different payload ranges selected using the range slider, show that:

1. The full payload range plot shows a visible gap around 4000 kg, so the data can be divided into two ranges: 0–4000 kg for low payloads and 4000–10000 kg for massive payloads
2. The 0–4000 kg range shows a higher launch success pattern than the massive payload range. In particular, the 3000–4000 kg range appears to have the largest success rate



Payload Mass vs Launch Outcome For All Sites

3. The 4000–10000 kg range has fewer successful launches, suggesting that massive payloads have a lower launch success rate than low payloads
4. The FT booster version appears to have the strongest launch success pattern among the booster types shown
5. Some booster types such as v1.0 and B5 have not been launched with massive payloads





Results

03 Predictive Analysis - Classification



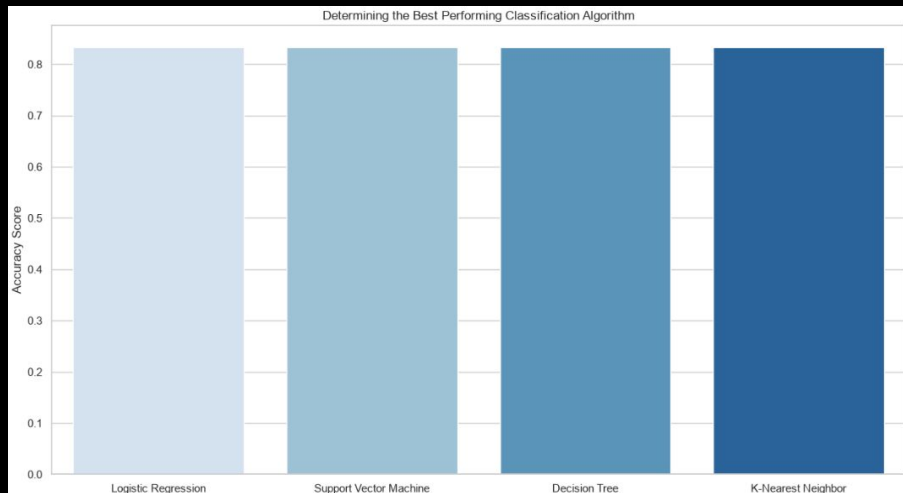
Classification Accuracy

The bar charts compare the Accuracy Score For All Classification Models and show that:

**All Classification Models achieved the same Accuracy Score of 83.33%.
Therefore, no single model outperforms the other.**

[124]:

	Algorithm	Accuracy Score
0	Logistic Regression	0.833333
1	Support Vector Machine	0.833333
2	Decision Tree	0.833333
3	K-Nearest Neighbor	0.833333



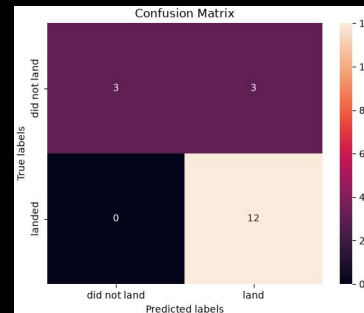
Confusion Matrix

As shown in the previous slide, all Classification Models achieved the same accuracy, so there is no single best-performing model.

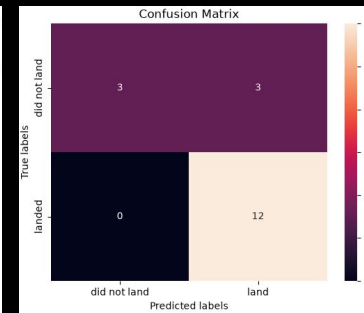
Their confusion matrices also produced the same results:

- 15 out of 18 test cases were classified correctly: 12 successful landings (True Positives) and 3 unsuccessful landings (True Negatives).
- 3 cases were misclassified as successful landings (False Positives), while no successful landings were incorrectly predicted as unsuccessful (False Negatives)

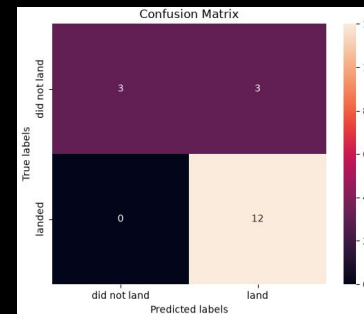
Logistic Regression



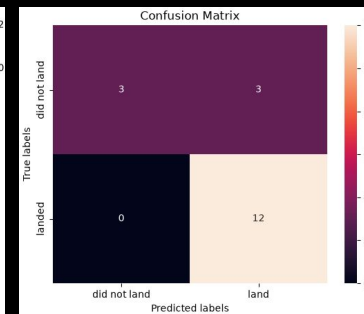
Support Vector Machine (SVM)



Decision Tree

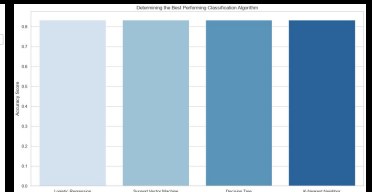
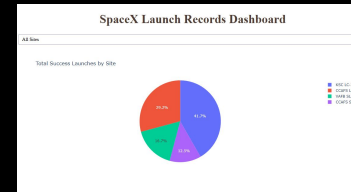
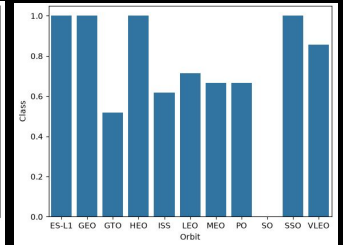
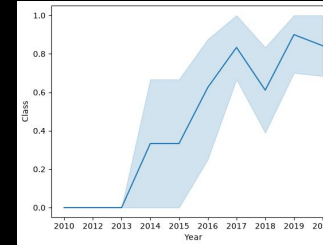


K-Nearest Neighbor (KNN)



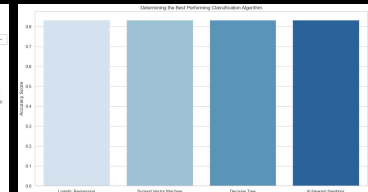
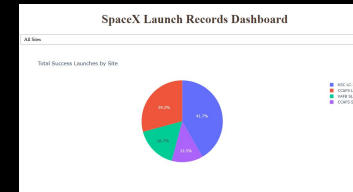
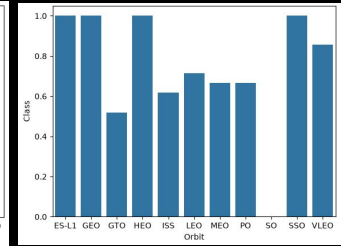
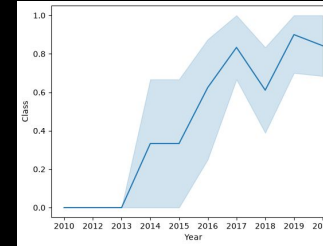
Conclusions

- Falcon 9 landing performance improved over time: higher flight numbers and later years show more successful landings, with success increasing after 2013 and most years after 2016 above 50%.
- Orbit type appears to be an important factor in landing outcomes. ES-L1, GEO, HEO, and SSO show the highest success rates at 100%, while SO shows the lowest success rate at 0%.
- Launch site performance differs across sites. KSC LC-39A had the largest share of successful launches, accounting for 41.7% of total successful launches, and also had the highest success ratio at 76.9%.



Conclusions

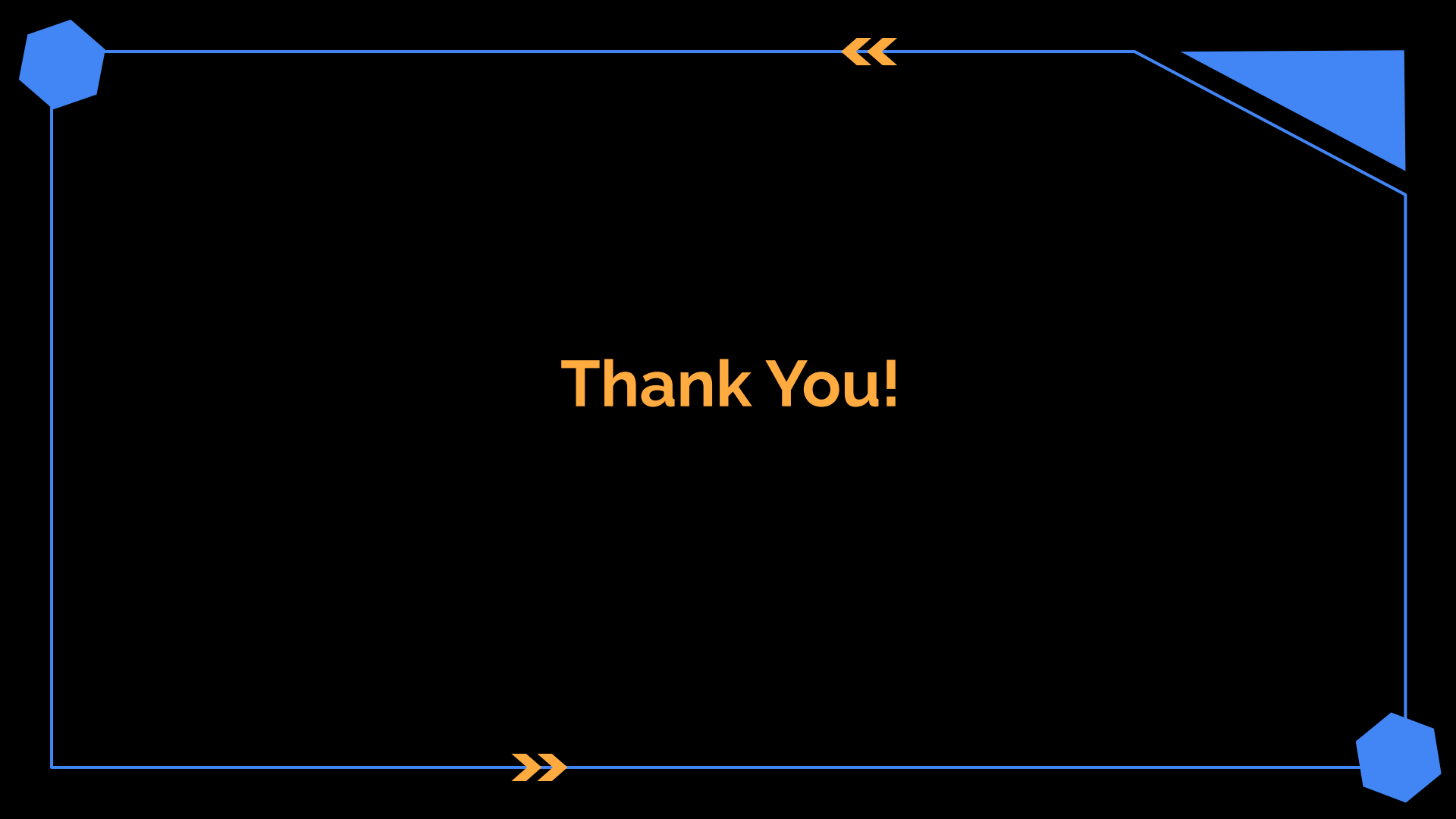
- Payload mass alone does not show a clear relationship with landing outcomes. However, dashboard results suggest that low payloads, especially the 3000–4000 kg range, show a stronger success pattern than massive payloads.
- All four classification models: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbor (KNN) achieved the same accuracy score of 83.33%, so no single model outperformed the others.





Appendix

- [Slide 7 Data Collection - SpaceX REST API Flowchart](#)
 - [Slide 8 Data Collection - Web Scraping Flowchart](#)
 - [Slide 9 Data Wrangling Flowchart](#)
 - [Slide 15 Predictive Analysis - Classification Flowchart](#)
 - [This Capstone's GitHub Repository](#)
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Thank You!